LESSON 10.1
EXPLORING RELATIONS

LESSON INTRODUCTION

MA.8.F.1.1 Given a set of ordered pairs, a table, a graph, or mapping diagram, determine whether the relationship is a function. Identify the domain and range of the relation.

LEARNING TARGETS

- I can recognize relations represented in different forms.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- What is a relation?
- How can you represent a relation?
- How do we refer to the input and output values in tables, mapping diagrams, and graphs?

LESSON NARRATIVE

Students explore input/output rules to develop an understanding of a relation as a set of input/output pairs. Relations are represented by mapping diagrams, tables, graphs, and sets of ordered pairs. Students connect input values to the independent variable, represented by x in ordered pairs, and output values to the dependent variable, represented by y in ordered pairs.

COMPONENT SUMMARY

10.1.1 WARM-UP *(5 MIN.)*

Create a pattern with numbers

10.1.2 EXPLORATION *(30-35 MIN.)*

Explore input-output pairs as relations in different representations

10.1.3 BRING IT TOGETHER *(5-10 MIN.)*

Recognize input values as the independent variables, represented by the x-values in ordered pairs, and output values as the dependent variables, represented by y-value in ordered pairs

10.1.4 WRAP-UP *(5 MIN.)*

Lesson reflection on the understanding of relations

RESOURCES

GLOSSARY

Key Terms:

- Relation

Additional Terms:

- Input
- Output

MATERIALS

- (Optional) Graph paper
- (Optional) Rulers
- (Optional) Template file
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

- Consider creating a template file with blank tables, graphs, and mapping diagrams to be handed out or shared electronically.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse the input and output values, independent and dependent variables, or x- and y-values.
- When graphing ordered pairs, students may struggle identifying the correct location of the point by mistakenly swapping the x- and y-value.

THINGS TO CONSIDER

- Consider creating reference cards or anchor charts with exemplar tables, mapping diagrams, and graphs.
- Make connections between the input value being the independent variable and the output value being the dependent variable because the output "depends" on the input.
- To set up future learning of functions, ask students to observe if any of the relations have more than one output for an input value.
- When creating mapping diagrams, list the values in each oval in increasing order, from top to bottom.

LITERACY AND LANGUAGE ROUTINES

Think Pair Share

10.1.2 Exploration asks students to determine a rule for an input/output table. Ask students to first think of a possible rule and then share their rule with their partner. Students determine if their partner's rule works for the given values and then correct any faulty rules.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

In 10.1.2 Exploration and 10.1.3 Bring It Together, students learn to represent relations using input/output tables, as mapping diagrams, graphs, and sets of ordered pairs. Students identify the input/output values, independent and dependent variables, and x- and y-values in each representation.

DIFFERENTIATE INSTRUCTION

Provide students with a template with blank tables, mapping diagrams, and graphs. This will avoid students spending time struggling to draw these structures and allow students to focus on developing the knowledge and skills of this lesson.

Having students first think about the questions independently and then discussing the questions with a partner will provide an auditory entry point. Provide a visual entry point by asking students to notice the similarities and differences between the representations of the relations. When plotting ordered pairs, provide a kinesthetic entry point by having students run their finger along the x-axis to the x-value and along the y-axis to the y-value and finally go the intersection of these points.

Enlarging the size of the tables, mapping diagrams, and especially the coordinate plane for the graphs may help students with visual or processing impairments when representing the relations. The larger size tables, mapping diagrams, and graphs can be made on a white board, chart paper, or (physical or digital) worksheet.

10.1.1 WARM-UP: CREATE A PATTERN

TEACHER MOVES

The pattern by which the numbers are changing can be an increasing pattern or a decreasing pattern.

DIFFERENTIATE INSTRUCTION

If students need different entry points, consider giving the first number in the pattern or the first two numbers in the pattern.

10.1.1 Warm-Up: Create a Pattern

1. Place a digit in each blue box so that the numbers change by the same amount each time. Use only digits 0 to 9, at most one time each.

$\square\square, \square\square, \square\square, \square\square, \ldots$

Answers will vary; example answer: $19, 28, 37, 46$

10.1.2 Exploration: Representing Mathematical Rules and Relations

An input-output rule is a rule that takes an input value and uses it to determine an output value.

input ⇨ $\boxed{\text{rule}}$ ⇨ output

For example, in the table shown, a rule was used to get each output value from its corresponding input value.

Input	-5	$-\frac{1}{3}$	0	2.5	6
Output	-15	-1	0	7.5	18

1. Work with your partner to determine the rule for the previous table. Then, complete the statement to describe the rule.

   Each [○ input / ○ **output**] value is the result of taking the [○ **input** / ○ output] and [○ adding 10 to it. / ○ dividing it by 3. / ○ **multiplying it by 3.** / ○ subtracting 10 from it.]

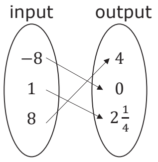
2. Work with your partner to complete the input-output table for each of the following rules. The first row has been completed as an example.

Rule	Table of Values
input ⇨ [add 1, then multiply by 4] ⇨ output	Input: $\frac{3}{4}$, 2.35, 42, -1; Output: 7, 13.4, 172, 0
x ⇨ [write 1 if the input is odd; write 0 if the input is even] ⇨ y	x: -12, -5, 6, 19; y: 0, 1, 0, 1

Each of the tables represent a ***relation***.

A ***relation*** is a set of input-output pairs.

Relations can also be represented by mapping diagrams. The mapping diagram below shows input-output pairs for the rule: "divide by 4, then add 2."



3. Work with your partner to complete the statements.

   In a mapping diagram an [○ **input** / ○ output] is mapped to its corresponding [○ input / ○ **output**] using an arrow. In the previous mapping diagram above, the input 8 results in an output of [○ **4** / ○ 0 / ○ $2\frac{1}{4}$] for the relation shown.

10.1.2 EXPLORATION: REPRESENTING MATHEMATICAL RULES AND RELATIONS

DIFFERENTIATE INSTRUCTION

Think-Pair-Share

Consider providing multiple entry points for learning with a Think-Pair-Share routine during the Exploration.

- Prompt students to independently make visual connections between the input and output values.
- Have students share their findings with their peers, making use of verbal processing.
- Direct students to then work with that partner to determine the rule before sharing and comparing responses as a whole group.

TEACHER MOVES

If the rule students wrote does not match with one of the statements in the third bubble of question 1, ask students which statement produces the outputs for the given inputs in the table.

DIFFERENTIATE INSTRUCTION

Students who struggle with fractions or decimals may benefit from the use of a calculator. Have students write and say each calculation they enter in the calculator to reinforce their computational skills.

ENRICHMENT

MA.K12.MTR.2.1: Multiple Representations

Position students to make connections between the representations and their uses.

- How are a table and a mapping diagram the same?
- How are they different?
- When might you use each? • What do you notice about the order of the numbers in each?

10.1.2 EXPLORATION: REPRESENTING MATHEMATICAL RULES AND RELATIONS (CONTINUED)

ENRICHMENT

Why do some of the numbers in the input oval of the mapping diagram have more than one arrow drawn?

QUESTIONING

For the second rule in the table, have students write the rule for the input and output as an algebraic expression. Students should consider the order of operations and grouping symbols. Expressions should be verified by substituting in the input value for the variable and simplifying.

TEACHER MOVES

Focus on the familiar word “depend” in question 5 and make connections when identifying which variables are independent and dependent.

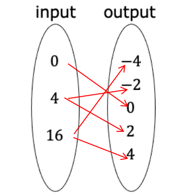
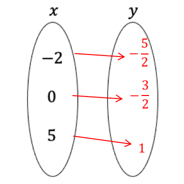
DIFFERENTIATE INSTRUCTION

When identifying the ordered pairs, provide a kinesthetic entry point by having students run their finger along the x-axis to the x-value and along the y-axis to the y-value and finally go to the intersection of these points.

DIFFERENTIATE INSTRUCTION

If students struggle with writing two observations, give them the option to write a wonder. For example, “I wonder if the number of arrows will always be the same as the number of points in the graph.”

4. Work with your partner to complete the mapping diagram for each of the following rules.

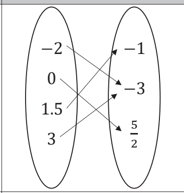
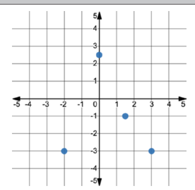
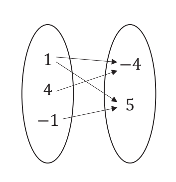
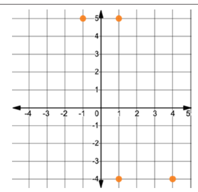
Rule	Mapping Diagram
input ⇨ find the square root(s) of the input ⇨ output	
x ⇨ subtract 3, then multiply by $\frac{1}{2}$ ⇨ y	

5. Complete the statements.

The ○ input values / ○ **output values** of a relation depend on the values used for the ○ **inputs.** / ○ outputs. Therefore, the inputs are the ○ dependent variables, / ○ **independent variables,** and the outputs are the ○ **dependent variables.** / ○ independent variables.

Relations can be represented using graphs.

For each relation below, its graph and mapping diagram are shown.

Relation	Mapping Diagram	Graph
Relation A		
Relation B		

6. With your partner, discuss your observations when comparing the mapping diagrams and graphs for each relation.
Write at least two observations you discussed.
Answers will vary. Possible responses include noticing that values do not repeat in mapping diagrams. Relation A does not have any repeating inputs/outputs while Relation B has more than one output for one of the inputs and more than one input that yields the same output.

7. Complete the statements.

   The mapping diagram for each of the relations has
   ○ 2
   ○ 3
   ○ 4
   input-output pairs indicated by arrows. The graph showing the same relation has
   ○ 2
   ○ 3
   ○ 4
   points.

   The input value from the mapping diagram matches the
   ○ x-value
   ○ y-value
   of each point on the graph, and its corresponding output value from the mapping diagram matches the
   ○ x-value
   ○ y-value
   of each point on the graph.

A set of ordered pairs can also be used to represent a relation.

8. The ordered pairs shown below can be used to represent Relation A from the previous table.

   $$\{(-2,-3),\left(0,\frac{5}{2}\right),(1.5,-1),(3,-3)\}$$

   Write the set of ordered pairs that represents Relation B from the previous table.

   $\{(1,-4),(1,5),(4,-4),(-1,5)\}$

10.1.2 EXPLORATION: REPRESENTING MATHEMATICAL RULES AND RELATIONS (CONTINUED)

ENRICHMENT

- Ask students to identify the similarities and differences between the representations of the relations.
- What are the similarities and differences between representing a relation in a mapping diagram, a graph, and a set of ordered pairs?

10.1.3 BRING IT TOGETHER: REPRESENTING RELATIONS

QUESTIONING

Ask students to create a relation that is aligned to a real-world context and represent it in a table, mapping diagram, and graph.

10.1.3 Bring it Together: Representing Relations

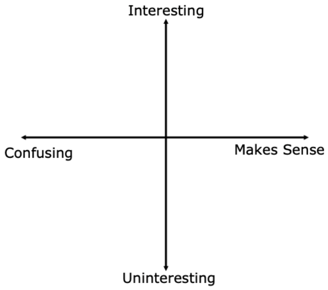
A relation is any set of input-output pairs. Relations can be represented using tables of values, mapping diagrams, a set of ordered pairs, or graphs.

1. Complete the statements.

   In a relation, the input values are the ○ dependent ○ independent variables. The input value is the ○ x-value ○ y-value of an ordered pair. The output values of a relation are the ○ dependent ○ independent variables. The output value is the ○ x-value ○ y-value of the relation. The relation can be represented as a set of ordered pairs or on a graph.

10.1.4 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest and understanding for today's lesson.
   Answers will vary.



10.1.4 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider providing one example for each quadrant followed by prompts that capture full use of the coordinate plane.

- Sample, Quadrant 1: "I felt excited about what I was learning, and I could explain it to someone else."
- Sample Prompt: Consider including a coordinate point to indicate more detail in your response.
- For example, (3, 3) might indicate moderate interest and sense making, while (3, 6) indicates the same level of interest but much more sense-making.